2022.M104A

2022L065A1EL



Coimisiún na Scrúduithe Stáit
State Examinations Commission

Leaving Certificate Examination 2022

Technology

Section A and Answerbook

Higher Level

Monday 27 June Afternoon 2:00 - 4:30

136 marks

Examination Number

Day and Month of Birth

For example, 3rd February is entered as 0302

Centre Stamp

The 2022 examination papers were adjusted to compensate for disruptions to learning due to COVID-19. This examination paper does not necessarily reflect the same structure and format as the examination papers of past or subsequent years.

Instructions

Write your Examination Number and your Day and Month of Birth in the boxes on the front cover.

Write your answers to all parts of the examination into this answerbook. This answerbook will be scanned and your work will be presented to an examiner on screen. Anything that you write outside of the answer areas may not be seen by the examiner.

Write your answers in blue or black pen. You may use pencil for sketches, graphs and diagrams only.

There are three sections to this examination. Attempt **all three** Sections.
Questions for Sections B and C are supplied separately but your answers must be written in this answerbook.

Section A 72 marks
Core – Short-answer questions.
Answer **any twelve** questions in this section.
Each question carries 6 marks.

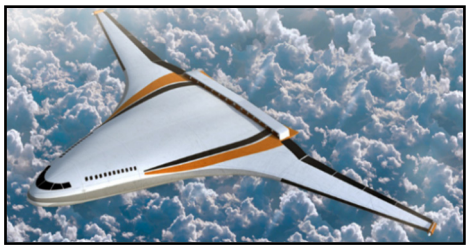
Section B 24 marks
Core – Long-answer questions.
Answer **one** of the two core questions presented.
Each question in Section B carries 24 marks.

Section C 40 marks
Options – Long-answer questions.
Answer **one** of the five optional questions presented.
Each question in Section C carries 40 marks.

Section A.

Answer **any twelve** questions. All questions carry 6 marks.
Write your answers in the spaces provided.

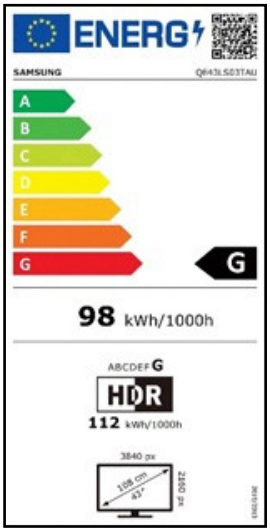
1. NASA are constantly striving to build efficiency into all phases in the design of aircraft, including development, manufacture and operation sequences.



Outline **two** challenges when trying to enhance the design of next generation aircraft.

1.
2.

2. On the 1st March 2021, a new rescaled energy label was introduced for products such as household refrigerators and freezers, washing machines, televisions, and electronic displays.



Outline **two** important pieces of energy information that the television label shown conveys to the consumer.

1.
2.

3. Outline an action for **each** of the following features of sustainable development, using the management of water resources as a theme:

Environmental:
Economic:
Social:



4. Since the start of 2021, Bus Eireann has introduced *hybrid* buses to their fleet and will continue to do so over the next few years.



(i) Explain what is meant by the term *hybrid*.

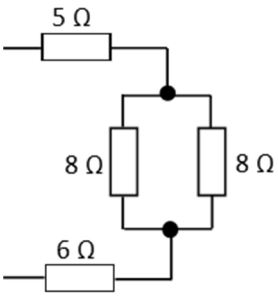
(ii) Outline **two** advantages of using hybrid buses over diesel-powered buses.

1.
2.

5. **(i)** Explain the function of the tolerance band on the resistor shown.



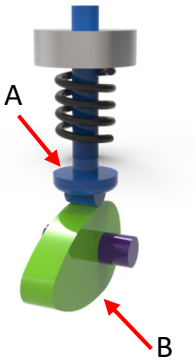
(ii) Calculate the total resistance of the circuit shown.



Calculation:

6. The graphic shows a mechanism which consists of two components, labelled **A** and **B**.

(i) Name component **A** and component **B**.



A:
B:

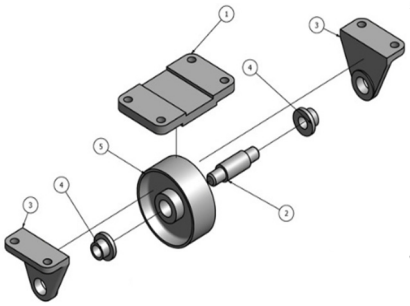
(ii) Outline the operation of the mechanism shown, referring to the movement changes and the spring.

7. The image shows a bridge under construction.
Compile a Gantt chart for the construction of a **typical** bridge.



8. An exploded view of a castor wheel is shown.

Sketch a pictorial view of the assembled castor wheel.



9. Two computer devices used to store data are shown.



(i) Explain the following terms:

SSD:
HDD:

(ii) Compare SSD and HDD in terms of *price*, *speed,* and *storage*.

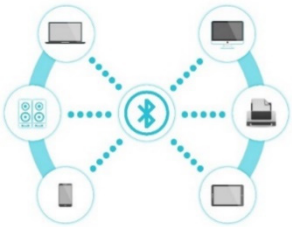
10. Risk assessment is a critical element in providing a safe environment in a school workshop. Complete the partial risk assessment guide, shown below, for the use of a soldering iron.

Risk Assessment – Technology Room – Soldering iron

Hazards	Is the hazard present? Y/N	What is the risk?	Risk rating H = High M = Medium L = Low	Control measures	Is this control in place? Y/N
Hot soldering iron	Y	Burns	H	Use soldering iron stand. Warning signs present.	Y Y
Damaged power lead	Y				
Fumes	Y				

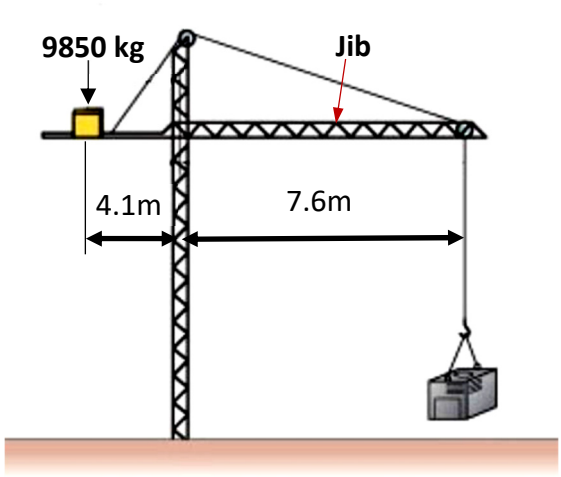
11. Briefly describe how Bluetooth technology can be used in **each** of the following tasks:

Data transfer:
Wire-free control:
Tethering:



12. The arm (jib) of a tower crane is 7.6 m in length. A counterweight of 9850 kg is placed at 4.1 m, as shown.

(i) Calculate the maximum load that can be lifted by the crane using this counterweight. (Ignore the mass of the beam.)

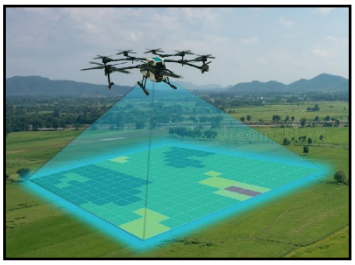


Calculation:

(ii) Explain what is meant by the term *triangulation*.

13. In recent years, data capture drones have become very popular.

(i) Name **one** application where drones could be used for data capture.



(ii) State **one** positive aspect and **one** negative aspect of using drones for data capture.

Positive:
Negative:

14. Planet Sustie is an Irish company that supplies 100% compostable, sustainable, and *biodegradable* partyware and tableware.

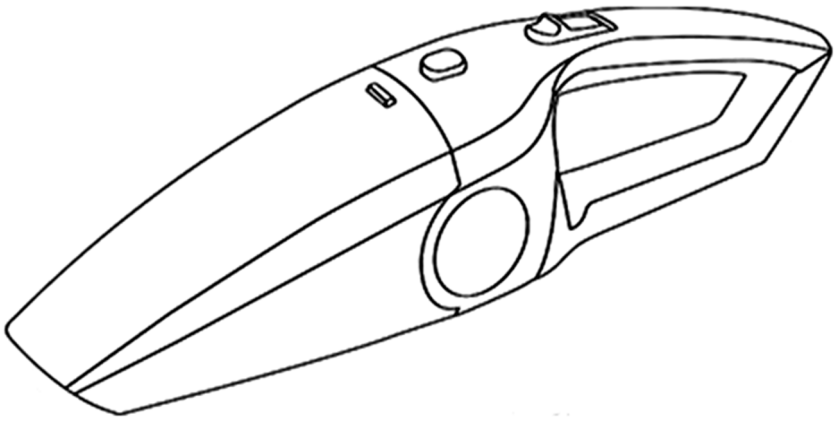
(i) Explain the term biodegradable.



(ii) Outline **one** advantage and **one** disadvantage of using biodegradable plates and cups.

Advantage:
Disadvantage:

15. Use **two** graphic techniques to enhance the representation of the handheld vacuum cleaner shown.



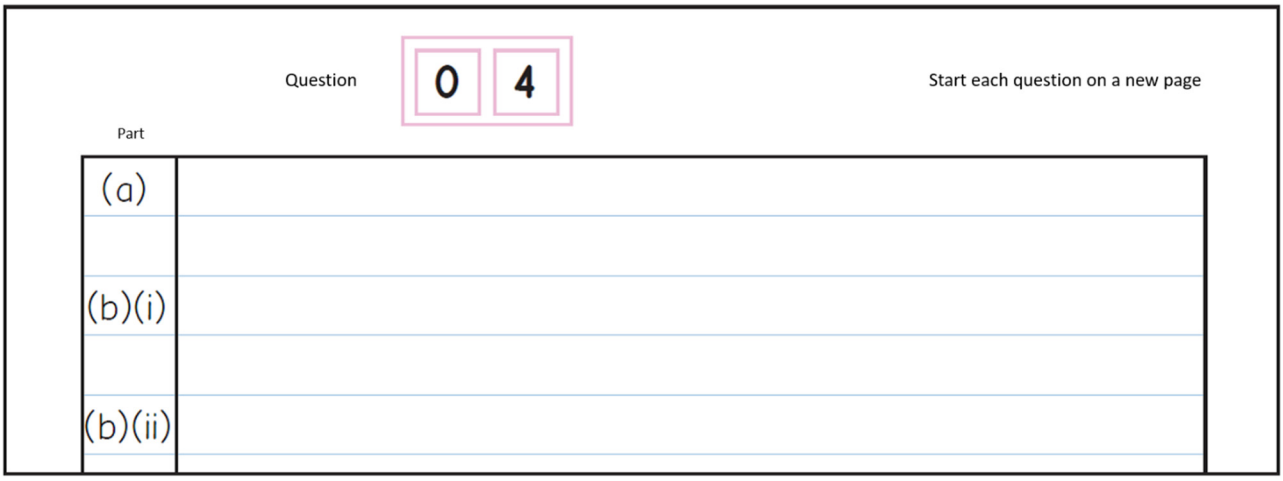
Answerbook for Sections B and C

Instructions

Questions for **Sections B** and **C** are supplied separately.

Start each question on a new page. Write the question number in the box at the top of each page.

Use the left-hand column to label each part, as shown below.



You do not need to use all of the pages in this answerbook. If you run out of space in this answerbook, you may ask the superintendent for more paper.

Write your answers in blue or black pen. You may use pencil for sketches, graphs, and diagrams only.

Question

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Acknowledgements

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Leaving Certificate – Higher Level

Technology Section A and Answerbook

Monday 27 June

Afternoon 2:00 - 4:30